Institutional Blockchain dApp Research

From landscape survey to a working credit vault on Rayls

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1. Executive Summary

The decision: build a curated collateral/liquidity vault for tokenized institutional assets — an ERC-4626 vault over a Morpho-Blue-style lending market — on **Rayls**. USDr deposits earn yield lent against AmFi tokenized-credit collateral, with an issuer-attested NAV oracle (Chainlink-delivered in production, availability on Rayls to be confirmed). The implementation is complete and audited locally (Section 6); deployment fallbacks, in order: Rayls testnet → Robinhood Chain.

Why this and not something else: banks are not asking for another chain — they are asking for compliant settlement, collateral, and asset-servicing workflows that plug into rails they already trust. Of the patterns proven to make independent builders real money (Section 3.2), the curated vault is the only one that is both institution-shaped and permissionlessly buildable. Rayls is the one venue where "institutional flows actually arriving" and "we can deploy now" intersect: Canton has the flows but is access-gated; Ethereum is open but saturated with curators; Rayls is live, EVM-compatible, has $100M of named collateral landing this month, and runs a $1M+ grant program. Its vault-distribution layer is claimed by a June 2026 partner cohort (Lagoon, Enzyme, Liqvid) — but the lending/borrow-market layer underneath, which such vaults route into on mature chains, remains publicly unclaimed. That is the wedge.

Layer map (three different games — don't confuse them):

- *Middleware* — Chainlink: we consume it, we don't compete with it.
- *Venues* — Rayls, Canton, Robinhood Chain, Ethereum: we deploy on one.
- *Products* — Ondo, Figure, Securitize, this vault: we compete here, and only here.

One-sentence thesis: institutional blockchain value concentrates in settlement, collateral, and compliance workflows — not in owning chains — so we build the first curated collateral vault on Rayls, the one live EVM venue with institutional collateral actually arriving, consuming interoperability middleware rather than competing with it.

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2. Scope & Methodology

Original brief: survey institutional/bank-facing blockchains — Chainlink, Canton, Polkadot, plus every other platform actually used by banks and credit institutions — map their L1/L2 architecture and the financial functionality built on top, identify a dApp use case with a genuine, underserved financial-institutional need, and build a proof-of-concept implementation (~80% research / 20% implementation).

Method: eight research rounds conducted 2026-07-15/16, covering ~65 platforms and products across ten categories (settlement rails, tokenized treasuries and funds, tokenized equities, stablecoins, CBDCs, private credit, commodities, real estate, custody infrastructure, institutional DeFi). A dedicated verification round re-checked every headline number against primary sources — issuer press releases, SEC filings, official dashboards — and corrected eight figures that circulated inaccurately in secondary reporting (among them Broadridge's monthly repo range, Ondo/BUIDL AUM, and Circle's charter type). Figures below that rest only on secondary reporting are flagged. Two subsequent strategy reviews — an adversarial pushback session and a no-deadline resequencing — were folded back into the conclusions (Sections 3.5 and 7).

Timeline context: research phase 2026-07-15 → 07-19, implementation 07-20/21, delivery 2026-07-22. The implementation was pulled forward and completed on 07-16 (Section 6).

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3. Market Landscape

3.1 Who owns the flows

Every large institutional flow is already owned by an incumbent or a consortium — and all of them are closed to outside builders.

Player	What they own	Scale (2026)
Broadridge (Canton tech)	Repo	$7–9T/month
JPMorgan Kinexys	Interbank payments	~$7B/day, targeting $10B
SWIFT + Chainlink CCIP	Cross-border messaging	11.5k banks, live
Circle + Tether	Stablecoin reserves	$2.75B revenue / $10B+ profit
mBridge (China-led)	Multi-CBDC cross-border	$55.5B cumulative, ~95% e-CNY
Deutsche Börse D7	Digital issuance registry	€44–90B, 45K issuances/week
Visa / Mastercard	Stablecoin settlement	$7B annualized, 9 chains
Euroclear / LSEG / SDX / HSBC Orion	EU digital bonds & CSDs	>CHF 2B / >$3.5B
Kalshi + Polymarket (ICE)	Prediction markets	$22–40B valuation / $2B ICE stake
Stripe Tempo + Circle Arc	Own payment L1s	Tempo live 03/2026; Arc testnet
DTCC + Nasdaq	US securities tokenization	Oct 2026 + Q4 2026 launches
Qivalis (37 EU banks)	Euro bank stablecoin	H2 2026 launch
Partior + Fnality	Bank settlement JVs	Live but narrow (secondary sources)

Conclusion: don't compete with rails — attach to them.

3.2 How independent builders actually make money

Proven patterns only — each row is backed by named, verified outcomes (References, Section 8).

Pattern	Proof (2026)	Revenue	Enterable?
Curated vault (Morpho curator)	Steakhouse $1.7B TVL; SocGen, Coinbase use it; a16z put $175M into Morpho	~$0.5M ARR	**YES — permissionless**
Fee layer on flow (Hyperliquid builder codes)	Phantom $20M/yr; pvp.trade $7.2M; $40M+ paid out total	$2–20M/yr	Yes, but retail-flavored
Grant capture (side income)	Rayls $1M+, Hedera $250K, Avalanche $30K	$10–250K one-off	Yes — pairs with any build
B2B plumbing (Securitize, Bridge)	Securitize $19.5M/qtr revenue, still a $7.9M net loss	$5–20M/yr	Slow — needs licenses
Retail vault on institutional collateral	Plume + ether.fi, $100M allocation, June 2026	Fee on AUM	Risky — incumbent already live
Own origination, go public (Figure)	$4.26B/qtr (8-K), ~$29B cumulative, NASDAQ-listed	Origination spread	No — must be a lender
Direct issuance, fee-on-AUM (Ondo, BUIDL)	~$2.1–2.4B AUM each	Compressing	No — BlackRock competes directly
Stablecoin / custody / CBDC issuance	Circle, Fireblocks, mBridge	—	No — confirmed closed
Failure mode — do not copy	Goldfinch: $18M defaults, ~70% real losses vs. 20% claimed	—	No

Conclusion: two viable small-team patterns exist — a fee layer on someone else's flow, or a curated vault. The vault is the institution-shaped one. Expect entry-level revenue (~$0.5M ARR range) as a curator; owning the protocol layer itself is the larger claim (Section 5).

3.3 Venue selection

Venue	Collateral on offer	Competition	Verdict
Rayls (the pick)	AmFi $100M July → $1B by mid-2027; USDr live	Vault layer claimed; **lending layer open**	Live since 04/2026, EVM, $1M grant program; confirm lending slot privately
Robinhood Chain	Stock Tokens, USDG	1 curator	Retail-flavored; open regulatory probe; launch-inflated TVL
Ethereum (Morpho)	Everything	5+ established curators	No differentiation for a newcomer
Canton	Broadridge repo, JPMD	Few apps	Best long-term economics; access-gated (see 3.5)
Hedera	Lloyds/Archax ~$10B flow	Thin DeFi	$250K grant, but HBAR captures no fee value
Avalanche Evergreen	FX/swaps (Spruce)	None	Still testnet — no production collateral to serve
Plume	BlackRock/Fidelity wrappers	ether.fi	Target niche already taken

Conclusion: build on Rayls. Fallbacks in order: Rayls testnet → Robinhood Chain.

3.4 Platform scorecard

Final position for every platform in the original brief (Chainlink, Canton, Polkadot) plus every other institutional chain surveyed:

Platform	What it really is	Verdict
Rayls	3-layer EVM institutional chain (Section 4)	**The pick** — live, $1M+ program, no lending incumbent
Chainlink	Oracle/interoperability layer, not a chain	Build *on top of* it — the vault's production oracle
Canton	Daml network, privacy-first; 62% of token issuance to app builders	Right economics; gate is sponsorship/curation, not language (3.5)
Polkadot	Relay chain + parachains	Ruled out — TVL ~$40M; its one institutional anchor (Centrifuge) left for EVM
Hedera	Public hashgraph, council-governed	Real ~$10B RWA flow, but the token captures no fees
Avalanche Evergreen	Permissioned EVM subnets	No production collateral yet (testnet)
XRP Ledger	Public payments L1	Watch RLUSD ($1.33B), not XRP itself
R3 Corda	Permissioned DLT, no token	Pivoting to Solana — the strongest signal in the table: even the most bank-native incumbent now believes distribution lives on public chains
Hyperledger Fabric	Permissioned framework	Dead end for outside builders — closed consortia
JPM Kinexys, Partior, Fnality	Banks' own closed rails	Proof the model works; not venues
SWIFT interlinking	Bridge from bank rails to DLTs	The rail we attach to via Chainlink CCIP

3.5 Corrections from the pushback review

Two challenges to the research were stress-tested and both updated the picture:

1. **"BlackRock competes" is a tailwind, not a warning.** What the research disqualified is *direct issuance, fee-on-AUM* — minting and distributing your own tokenized fund against BlackRock's brand on compressing fees. The chosen pattern — curator **on top of** existing collateral, not issuer of new collateral — routes around that moat entirely. BlackRock/Ondo's presence validates the niche; it belongs in the pitch as demand evidence.
2. **Canton's real gate is curation, not the Daml language.** Free LocalNet, quickstart scaffolds, and AI-assisted Daml tooling have eroded the skill barrier; the true gate is **validator sponsorship plus "featured app" status** — granted by Digital Asset, and the thing that actually unlocks the 62% reward pool. That is a relationship gate, the same shape as the Rayls situation, just with a different counterparty. **New finding: Zenith, an EVM compatibility layer, is live on Canton.** If mature, the same Solidity vault + lending-market code built for Rayls plausibly redeploys to Canton at low switching cost, and Canton's pre-built collateral/financing modules could shortcut the repo concept (Section 5.4) far more than the original "phase 2, months away" framing assumed. Requires a hands-on technical spike before trusting the marketing copy (Section 7, step 4).

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4. The Venue: Rayls

Architecture — three layers:

1. **Privacy Nodes** — a private EVM chain each institution runs on-prem or in private cloud, for tokenizing and managing assets internally.
2. **Private Networks** — permissioned hub-and-spoke interconnect for institution-to-institution transfers.
3. **Public Chain** — an EVM-compatible public L1, live since 2026-04-30, connecting the private layers to open DeFi.

Built on Reth with periodic state-root anchoring to Ethereum. RBFT consensus, migrating to **Axyl** in 2026 (~250k TPS claimed — internal prototype benchmark, unaudited). Gas is USD-pegged and paid in **USDr**, Rayls' native stablecoin (predictable institutional operating costs). MEV protection and native KYC/KYB identity layers are protocol features. **Enygma** — the privacy protocol (ZK proofs + post-quantum cryptography, selective disclosure for regulators) — reaches the public chain in Q3 2026. Assets move from Privacy Nodes to the Public Chain via a lock-and-mirror bridge (`teleportToPublicChain()` locks on the private side; a relayer mints a mirror token publicly) — likely how AmFi's tokens arrive. Public-chain access is operator-gated by design.

RLS token: 10B fixed supply, no minting. Every transaction fee: 50% burned, 50% to the validator Network Security Pool. TGE 2025-12-01 (~15% circulating; remainder unlocks Dec 2026 → 2029); listed on Coinbase, Gate, MEXC. Distribution: 50% community/ecosystem, 22% investors, 17% contributors, 11% technology provider — non-community allocations on 4-year vesting.

Traction: Núclea (Brazil's largest financial-market infrastructure, 31B transactions/yr) and Cielo (largest card acquirer) live per Rayls; core settlement infrastructure for Brazil's **Drex** CBDC pilot; 25+ institutions claimed. Independently notable: **JPMorgan Kinexys deployed on the network**; **Tether made a strategic investment in Parfin** (Rayls' core developer); **Mastercard added Rayls to its Crypto Partner Program** (March 2026); XP (Brazil's largest retail broker) on the asset side; Parfin's own FX desk migrating ~$400M/month of flows from Q3 2026. Team: ex-BTG Pactual executives plus leads from Siemens, J.P. Morgan DLT, State Street, and AWS; backed by ParaFi, Framework, Valor, and Borderless.

Read critically: the litepaper is marketing material — the TPS figure is an unaudited prototype number and several "live in production" claims are not independently verified. The load-bearing facts above (mainnet date, TGE, the AmFi commitment, Kinexys/Tether/Mastercard) were re-verified against primary sources during the verification round.

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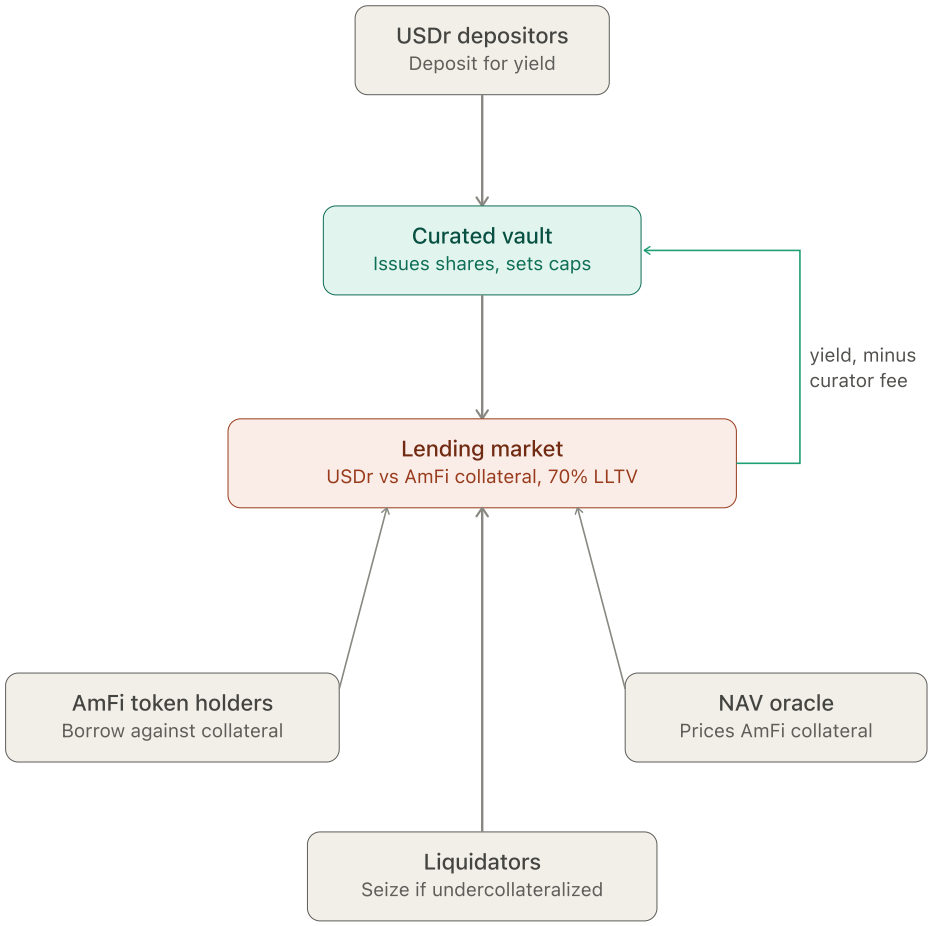
5. Product Design: A Curated Credit Vault

5.1 The mechanic

A **lending market** is a pool with two assets and one rule. Lenders deposit USDr to earn interest; borrowers lock AmFi tokenized-credit tokens as collateral and borrow USDr against them, overcollateralized (70% liquidation LTV in the PoC). If collateral value falls below the threshold, anyone may permissionlessly **liquidate**: repay the debt and seize collateral at a discount (5% bonus). Why would an institution borrow

against tokenized credit instead of selling it? The same reason repo exists in TradFi — cash liquidity without giving up the yield-bearing asset. A holder of receivables yielding ~14% who can borrow at ~8% keeps the spread and redeploys the cash.

A **curated vault** (ERC-4626) sits on top so depositors never touch raw markets: deposit USDr, receive vault shares; the curator decides allocation across markets and caps; depositors get one-click, risk-managed yield; the curator earns a performance fee (15% of yield in the PoC — the rate Steakhouse actually charges on Morpho).



Example: $100 AmFi collateral supports $70 USDr borrowed

Why Rayls specifically, not Ethereum:

1. **Compliance is native, not bolted on.** Rayls has a protocol-level identity layer, so KYC-gated markets don't need a third-party compliance stack — the SG-FORGE-on-Morpho precedent (a regulated bank's stablecoins inside permissionless DeFi) with less friction.
2. **First mover = protocol owner.** On Ethereum this build would be curator #6 (~$0.5M ARR territory). On Rayls nothing like it exists yet — a shot at owning the *protocol* layer, which Morpho's investors valued at ~$2B on a mature chain.
3. **Roadmap stacking.** This vault's liquidity is the cash leg a future repo/collateral-mobility product needs — and repo is the single biggest proven flow in the entire survey ($7–9T/month at Broadridge).

5.2 The collateral: AmFi tokenized credit

AmFi is Brazil's leading private-credit tokenization platform — regulated under the Central Bank of Brazil and the CVM, assets held by licensed custodians, ~R$3B (~$500M+) issued to date. In December 2025, AmFi and Rayls announced a partnership migrating AmFi's portfolio onto Rayls: **$100M by July 2026 → $500M by January 2027 → $1B+ by June 2027.**

An AmFi token is an on-chain claim on a pool of Brazilian private credit — receivables financing, structured credit notes, FIDC-style senior/junior tranches. Its value tracks the pool's NAV (principal + accrued interest − defaults), and cash flows come from real businesses repaying real loans. That is exactly why it works as collateral (real yield, real NAV) and exactly why it carries real credit risk.

5.3 Risk analysis

- **Oracle.** Tokenized credit has no second-by-second market price. Production needs an issuer-attested NAV feed (Chainlink-delivered if available on Rayls) and conservative LTVs to absorb valuation lag. The implementation enforces a non-zero-NAV invariant so a broken oracle fails loudly instead of silently mispricing collateral.
- **Credit risk does not vanish.** Underlying receivables can sour — Goldfinch lost ~70% of depositor funds (vs. a claimed 20%) on unenforceable credit. Defense: overcollateralization, conservative caps, and never pretending the collateral is risk-free.
- **Liquidation depends on an exit.** Liquidation is permissionless and the discount is the entire incentive — but on Ethereum a liquidator instantly sells seized ETH on a DEX, and AmFi tokens have no equivalent. Three checks, three dead ends: AmFi's own secondary marketplace has said "coming soon" since 2023; no AmFi/USDr pool exists on Rayls' Algebra DEX; AmFi's originator-mediated model makes the tokens buy-and-hold by construction. **The fix — matching how AmFi's real exit mechanics work — is an issuer-redemption pool rather than an open-market sale.** This is built into the implementation, and it doubles as a second product opportunity: solving the redemption gap is what makes RWA collateral usable at all, and nobody else is positioned to claim it either.

5.4 Alternatives considered

Concept	Copies	Why not now
Intraday repo / collateral mobility	Broadridge DLR ($7–9T/month), DTCC Collateral AppChain, Canton repo	Needs bank counterparties and months of onboarding — phase 2, with vault liquidity as its cash leg; outreach starts now (Section 7)
LatAm stablecoin/payment orchestration	Stripe/Bridge ($1.1B acquisition), Visa rails ($7B annualized)	Sits on Rayls' own USDr roadmap — joint product or conflict; requires an explicit partner conversation first
Institutional RWA-perps venue	Hyperliquid (~$700–840M/yr revenue, 44% of RWA-perp volume)	Highest ceiling in the survey, but needs deep liquidity and a real risk engine; phase 3 — the lending market's rate curves and NAV outputs are designed to become its price feed

None of these were buildable in the implementation window; all three are roadmap, not rejects.

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6. Implementation Status

The vault is built — it is **this repository** (the research document ships with the code it motivated). Built ahead of the nominal window to support the scheduled demo/walkthrough (2026-07-22).

Contracts (`src/`):

Contract	Role
`MockUSDr.sol`	Stand-in for Rayls' USDr gas stablecoin (6 decimals)
`MockAmFiCredit.sol`	Stand-in AmFi credit token: owner-set NAV (oracle stand-in) + **issuer-redemption pool** (the Section 5.3 exit fix)
`LendingMarket.sol`	Morpho-Blue-inspired market: overcollateralized borrowing, utilization-based rates, permissionless liquidation, virtual-share donation-attack protection, **bad-debt socialization**
`CuratedCreditVault.sol`	ERC-4626 curator vault: capped allocation, automatic liquidity pull on withdrawals, EIP-4626-compliant `maxWithdraw` / `maxRedeem` under market illiquidity

Verified end-to-end on a local Anvil chain: the narrated demo (`script/Demo.s.sol`) runs deposit → curator allocation → borrow → NAV writedown → liquidation → issuer redemption, with final state independently cross-checked via `cast` against real on-chain data.

Pre-audit review: a three-reviewer adversarial pass found and fixed two HIGH-severity issues — (1) liquidations that exhausted a borrower's collateral left unbacked bad debt accruing phantom interest with no write-off (fixed with Morpho-style pro-rata socialization); (2) the vault's `totalAssets()` projection ignored curator-fee dilution, which could revert full withdrawals and leak value between depositors (fixed to project fee dilution exactly) — plus medium fixes (EIP-4626 `max*` compliance under illiquidity; interest forfeiture on small markets) and hardening (25% fee cap, zero-NAV oracle guard, 18-decimal collateral assertion).

Current state: 63 tests across 4 suites, all passing, including a regression test for every review finding. Coverage: vault 100% lines/branches; lending market 97% lines / 81% branches. **Not yet done:** deployment to the real Rayls testnet — blocked solely on gated network credentials (RPC, chain ID, registry address, operator approval) expected via the partnership, not on anything technical.

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7. Action Plan & Roadmap

Steps 1–5 are conversations and checks — cheap, no code, start immediately; 1 and 5 gate the biggest decisions. Step 6 is calendar-gated, so it starts now even though it feels like "later." Steps 7–8 carry the engineering weight.

1. **Ask Rayls the bundled roadmap question** (highest-leverage single action): (a) is a lending/borrow-market layer already privately committed to someone, (b) would LatAm payment orchestration be a joint product or a conflict, (c) what does the $1M developer program's application route require. Same channel: the gated Public Chain credentials — the only thing blocking testnet deployment.
2. **Verify AmFi token mechanics directly with AmFi** — token standard, transferability vs. KYC-gating, on-chain NAV publication, and the *actual* redemption path for a liquidator holding seized tokens. Kills

or reshapes the collateral leg; needs a direct answer.

3. **Confirm Rayls deployment specifics** — RPC/chain ID/faucet, permissionless vs. Predicate-gated deployment, and Chainlink oracle availability (else a fallback NAV source is required before any market can price collateral).
4. **Run a technical spike on Zenith** (Solidity-on-Canton) — deploy something trivial and see what breaks; determines whether this codebase has a low-cost second home on Canton.
5. **Open a Canton access conversation** — validator-sponsorship timeline and what "featured app" status (the 62%-pool unlock) actually requires. A relationship gate: the sooner it starts, the sooner Canton is a Year-1 target or ruled out.
6. **Start bank-partner outreach for the repo concept now, not after the vault ships** — its bottleneck is bank-onboarding calendar time, which ticks independently of build progress.
7. **Build the lending market as a real product, not a demo** — multi-market support (AmFi now; Nimofast and deposit tokens as they land), real oracle integration, a proper curator interface, issuer-redemption liquidation tooling — with rate-curve/NAV outputs designed to double as the future perps price feed.
8. **Re-sync checkpoint** once steps 1 and 5 return answers — re-rank everything before new resources move.

Sequencing principle: blockers split into *engineering-time* gates (solved by more build days) and *calendar/relationship* gates (solved only by starting conversations earlier). The vault stays the anchor; repo moves up as a parallel relationship track; payment orchestration folds into the step-1 conversation; perps becomes an architectural constraint on the vault's data outputs. **Run "vault → repo → flow" as overlapping tracks, not a sequence.**

Roadmap:

- **Now:** lending vault on Rayls — proven pattern, working code, grant-eligible.
- **Quarter 1:** compliant lending markets for every asset entering Rayls; bank conversations for repo running in parallel.
- **Year 1:** repo/collateral mobility with the bank pipeline, vault pools as the cash leg.
- **The line:** "*we're building the credit & collateral layer of Rayls.*"

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8. References

Primary and load-bearing sources per claim. Where only secondary reporting exists, it is noted in the text.

Flow owners (3.1)

- Broadridge DLR monthly volumes: Dec ~$9T · Mar +392% YoY · Jun $7.5T
- JPMorgan Kinexys ~$7B/day: DL News · JPMorgan
- SWIFT + Chainlink CCIP: Chainlink Q1 2026 review
- Circle FY25 ($2.75B revenue, $70M net loss): Circle; Tether $10B+ 2025 profit: Tether
- mBridge $55.5B: The Block / Reuters
- Deutsche Börse D7: WFE/Clearstream

- Visa $7B annualized across 9 chains: The Block
- SDX >CHF 2B: SIX; HSBC Orion >$3.5B + UK DIGIT mandate: HSBC
- Kalshi $22B valuation / ~$260M 2025 revenue: Fortune; Polymarket / ICE $2B: company.gi
- Stripe Tempo ($500M @ $5B; mainnet 03/2026): Fortune
- DTCC tokenization (Oct 2026) + Chainlink Collateral AppChain (Q4 2026): DTCC
- Qivalis 37 banks / 15 countries: Piraeus

Money patterns (3.2)

- Morpho $175M round (a16z/Paradigm/Ribbit, ~$2B; $11B+ deposits): Fortune · CoinDesk
- Steakhouse Financial ($1.7B TVL, ~$0.5M ARR, 15% fee): DefiLlama · Morpho
- Hyperliquid builder codes (Phantom $20M, $40M+ total): Hyperliquid · CoinGecko Research
- Securitize Q1 2026 ($19.48M revenue, $7.93M net loss): StockTitan
- Plume + ether.fi $100M vault: PRNewswire
- Figure Q2 2026 ($4.26B quarterly volume, ~$29B cumulative): GlobeNewswire
- Ondo / BUIDL AUM: Eco fund comparison
- Goldfinch wind-down (~$18M defaults, ~70% real losses): The Defiant · DL News

Venues & platforms (3.3–3.4, 4)

- Rayls mainnet (2026-04-30), USDr, ecosystem: Rayls · Rayls — what's next
- Rayls $1M developer program: Rayls
- Rayls × AmFi partnership ($100M → $1B): GlobeNewswire
- Rayls × Lagoon vault cohort: Rayls
- Canton app-reward economics (62% to apps): Canton Coin whitepaper
- Robinhood Chain mainnet (07/2026): Robinhood · Arbitrum
- Hedera ~$10B RWA settlement (ecosystem reporting): Hedera case study
- R3 → Solana pivot: CoinDesk; Centrifuge leaves Polkadot: Centrifuge governance
- US CBDC ban through 2030: CoinDesk; Circle national trust bank charter: Circle